

Department of Economics

Jahangirnagar University

M.Sc. Final Examination 2022

Course Title : Environmental Economics

ECON: 506

Marks- 70

Time- 4 Hours

Answer any Five the following Questions. All parts of a question must be answered consecutively.

✓✓

What is the main concern of Utilitarianism?

✓✓

Critically explain the concept of preference-satisfaction utilitarianism as a basis for modern welfare economics.

✓✓

What are Utilitarianism's weaknesses? How Rawl's 'Theory of Justice' does challenges classical utilitarianism? Explain.

3+5+6

2. a) Evaluate the empirical evidence for the Environmental Kuznet's Curve (EKC) hypothesis.
b) Discuss the policy implications of the EKC including the potential for decoupling economic growth from environmental degradation.

6+8

✓✓✓

3. Distinguish between flow and stock pollutants. Analyze efficient levels of flow pollutants and stock pollutants.

✓✓ Explain the role of spatial differentiation for emission targets

✓✓ Find the efficient solution (M^* and A^*) for uniformly mixed and short-lived stock pollutant with the following information:

$$\text{Total Benefit, } B = \begin{cases} 96M - 0.2M^2 & \text{for } 0 \leq M \leq 240 \\ 11520 & \text{for } M > 240 \end{cases}$$

$$\text{Total Damage, } D = 0.2A^2 \text{ for } A \geq 0$$

and in steady state assume that $A = 2M$

where,

M is the Emissions flow

A is Stock Pollutant

6+4+4

✓✓

4. What are the environmental consideration for risk assessment? Explain the risk assessment process regarding environmental degradation.

✓✓ What is risk matrix? How much environmental risk is acceptable?

✓✓ Explain various tools for assessing/mitigating risk.

7+3+4

- ✓ a) Define compensating variation (CV) and equivalent variation (EV) in consumer preference. Show the relation among CV, EV and Marshallian Consumer Surplus (MCS).
- ✓ b) Suppose that an individual has the utility function $U = E^{0.25} + Y^{0.75}$ where E is the some index of environmental quality and Y is income. From an initial situation where $E = 1$ and $Y = 100$, Calculate Compensating Surplus (CS) and Equivalent surplus (ES) for an increase in E to the level 2 and for a decrease in E to the level 0.5.
- ✓ c) What is the difference between direct and indirect method of environmental valuation?

6+5+3

6. a) What are some examples of ways in which climate change will affect people around the world unequally? What are the effects of this inequality?
- b) Should the world focus efforts on mitigation or adaptation? Why?
- c) What is NDC? What are the benefits of letting each country decide its own NDC? What are the limitation of this approach? Why do you think negotiators choose this approach rather than negotiating each country's responsibility as they did in the Kyoto Protocol?

6+3+5

- ✓ a) What is the CVM method of environmental valuation? In which regards Contingent valuation method can be considered superior to all other techniques for valuing the environment?
- ✓ b) What is the Travel cost method of valuing environment? How the Travel cost method is an example of revealed preference? Explain the limitations of the Travel cost method.

7+7

8. Write short notes on: Any four
- Option price and option value
 - Certainty Equivalence and Risk premium
 - Residual risk
 - Net zero emission.
 - Paris Treaty
 - Irreversibility and Uncertainty.
 - Loss and Damage fund

3.5 x 4 = 14

- ✓ a) Explain the concept of Utilitarianism? What are the main problems with utilitarianism? (5)
- b) What are the principles of Rawls "Theory of Justice"? How does Rawl's 'Theory of Justice' challenge classical utilitarianism? Explain. (6)
- c) Does economic growth inevitably lead to environmental degradation? (3)
- ✓ a) Consider a producer to consumer externality as

$$U^A = U^A(X^A, Y^A, S) \text{ with } \frac{\delta U^A}{\delta S} < 0$$

$$U^B = U^B(X^B, Y^B, S) \text{ with } \frac{\delta U^B}{\delta S} < 0$$

$$X = X(K^X, L^X)$$

$$Y = Y(K^Y, L^Y, S) \text{ with } \frac{\delta Y}{\delta S} > 0$$

What would be the suitable tax rate to correct the externality? Explain with appropriate graph. (8)

b) Suppose that a wood pulp mill is situated on a bank of a river. The marginal benefits and costs are as follows-

$$MC = 10 + 0.5Y$$

$$MB = 30 - 0.5Y$$

Where Y is tons of wood pulp produced. Each ton of wood pulp produces pollutant flows into the river which cause damage valued at \$10. (6)

- Find the socially efficient output level of wood pulp and the private profit maximizing output level. Why are these two levels different?
 - Explain why the socially efficient output of wood pulp is lower than the private profit maximizing output level.
 - How large would marginal external cost have to be for it to be socially desirable that no wood pulp is produced?
3. a) What is the environmental risk? Distinguish between risk and uncertainty. (3)
- b) Explain the risk management process regarding environmental degradation. (5)

c) Show how risk, irreversibility and uncertainty affect environmental decision making. Suppose

$MNB_1 = 10 - \left(\frac{A_1}{2}\right)$ and $MNB_2 = 20 - \left(\frac{A_2}{2}\right)$ known with certainty, find the levels of A_1 and A_2 (6)

i. if there is no irreversibility,

ii. if irreversibility applies but is ignored in decision making and

iii. if irreversibility is taken into account.

✓ a) Define compensating variation (CV) and equivalent variation (EV) in consumer preference. Show the relation among CV, EV and Marshallian Consumer Surplus (MCS). (7)

b) Suppose that an individual has the utility function $U = E^{0.25} + Y^{0.75}$ where E is some index of environmental quality and Y is income. From an initial situation where $E = 1$ and $Y = 100$, Calculate Compensating Surplus (CS) and Equivalent surplus (ES) for an increase in E to level 2 and for a decrease in E to level 0.5. (7)

✓ a) Explain the Travel Cost Method (TCM) and the typical problems encountered when using it. Give potential solutions for these problems and explain the difference between the Zonal Travel Cost Method and the Individual Travel Cost Method. (8)

b) How does revealed preference become an important instrument to assess Travel cost method for environmental valuation? (2)

c) Define and explain Hedonic Pricing technique for environmental valuation. Why is it widely used? (4)

6. a) What is the difference between Global Warming and Climate Change? (3)

b) Distinguish between Adaptation and Mitigation to the Climate Change. (4)

c) Is mitigation better than adaptation? Explain why Adaptation is more emphasized than Mitigation. (7)

7. a) Evaluate the empirical evidence for the Environmental Kuznets Curve (EKC) hypothesis. (6)

b) Discuss the policy implications of the EKC, including the potential for decoupling economic growth from environmental degradation. (8)

✓ Write Short Notes: Any four (3.5x 4 = 14)

✓ a) Option price and option value

✓ b) Contingent Valuation Method

b) Risk Matrix

g) Loss and Damage Fund

c) Environmental Kuznets Curve

h) Certainty Equivalence and Risk premium

✓ d) Net zero emission.

e) Paris Treaty